DISSOLVING AND DISSOCIATING 4

A general (but not infallible) rule says **like dissolves like**. This means that polar solvents tend to dissolve polar/ionic substances and non-polar solvents tend to dissolve non-polar substances.

Remember, in a solution:

- the solvent and solute particles are evenly distributed (homogenous) throughout the solution
- the solute particles are too small to be seen
- there are forces of attraction between solvent and solute particles.

Before a substance can dissolve in water the forces between its particles and between water molecules must be broken. This results in new bonds being formed between the water and the solute. Breaking bonds requires energy to be put in (endothermic; *endo*: within); making bonds gives energy out (exothermic; *exo*: outside). If a covalent molecule is very large (it can consist of tens of thousands of atoms, e.g. cellulose and polythene) or is non-polar, it will not dissolve in water. The bonding in covalent network structures (e.g. diamond and silicon dioxide) is too strong to be overcome by any possible bonds between particles in the structure and water molecules.

1 Suggest why a lattice network is so much stronger than other forms of bonding.

2 'A substance will only dissolve in a solvent if the attractive forces between solute and solvent particles are similar to those between the solvent/solvent and solute/solute particles.' Explain this statement in your own words.

3 Sodium chloride, NaCl, is soluble in water. Its structure is shown in figure 1.

a Is sodium chloride a covalent compound or an ionic compound? How do you know?

b Is water a polar solvent or a non-polar solvent? Explain your answer.

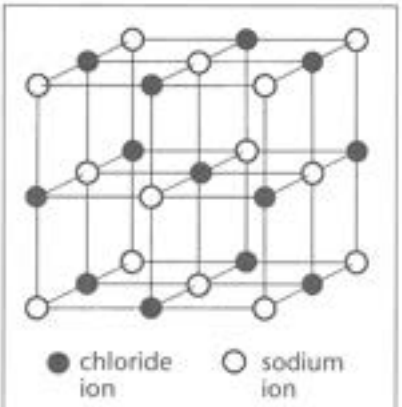


Figure 1

When positive and negative ions from an ionic lattice are separated, this process is called **dissociation** (*dissociate*, to part). The dissociated ions are surrounded by water molecules; they are **hydrated**. This can be shown by the equation:

$$NaCl\ (s) + H_2O\ (l) \rightarrow Na^+\ (aq) + Cl^-\ (aq)$$

4 Write an equation to show how each ionic solid dissolves and dissociates in water.

a potassium fluoride (KF) **b** magnesium chloride ($MgCl_2$) **c** potassium carbonate (K_2CO_3)

5 Not all ionic compounds are soluble. Why might an ionic compound be insoluble?

Sucrose ($C_{12}H_{22}O_{11}$, sugar) is a soluble covalent molecule. Although it is covalent, it does have polar parts which can be attracted by water molecules, forming hydrogen bonds. The dissolving of sugar in water can be represented by the equation:

$$C_{12}H_{22}O_{11}\ (s) + H_2O\ (l) \rightarrow C_{12}H_{22}O_{11}\ (aq)$$

6 Both sugar (sucrose) and common salt (sodium chloride) dissolve. How does the dissolving of sugar differ from that of sodium chloride?

Some covalent compounds are very highly polarised. Figure 2 shows the structure of hydrogen chloride. The hydrogen is relatively positive and the chlorine is relatively negative. The covalent bonds in polar molecules can be broken by the electrostatic forces between water and solute molecules. New bonds are formed. One example is when hydrogen chloride dissolves and ionises in water:

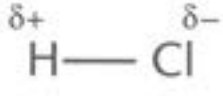


Figure 2

$$HCl\ (g) + H_2O\ (l) \rightarrow H_3O^+ + (aq) + Cl^-\ (aq)$$

(Note that H_3O^+ is known as the hydronium ion; it is responsible for acidic properties.)

7 What is an aqueous solution of hydrogen chloride called?

8 A similar reaction to that of hydrogen chloride dissolving in water occurs when liquid nitric acid (HNO_3) dissolves in water. Represent this in a balanced equation.

9 Give two reasons why a covalent molecule might not dissolve in water.

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